

MAE 3780 Robotics Competition

For this competition, we designed and built a moving robot in order to collect cubes in a 1V1 format. For this assignment my role was designing the code and wiring in order for the robot to function as intended. Our wiring involved a basic H-bridge motor that utilized an Arduino in order to drive the motors at specific intervals. For the competition we decided to go for a simplistic approach. Our code was simple: drive straight, turn 115 degrees, and drive straight again. This involved initiating the outputs in the Arduino, driving the motor by setting the Arduino pins to high, and then setting everything to zero in order to be an obstacle for the other robot.

As part of this assignment as well, I programmed a color sensor and QTI in order to meet the milestone requirement of border detection. This involved using various timers in order to consistently detect what color the sensor was seeing and utilizing detection thresholds for various functions. This involved setting the initial color, then once the detection threshold was met for the opposite color, the robot would turn around and drive until the back border. The process also involved the QTI sensor detecting the boards through a separate timer. Once the QTIs detected a black border, it would adjust or stop depending on whether the left, right, or both sensors detected black.

Within the competition we learned various debugging techniques, including using print functions and timer delays in order to find where we went wrong. The most challenging aspect of this project was being able to have the various elements all work in tandem. This proved a challenge; however, through debugging, we were able to get a working robot that functioned as intended.

